

Simulation Analysis: baseline-honest-no-drop-large-degree-20251024-131216

SIMULATION: baseline-honest-no-drop-large-degree-20251024-131216

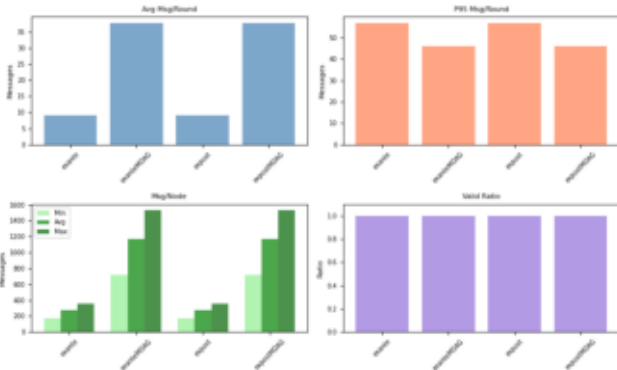
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Files: 1000, Records: 434638

Network Graph Analysis:
Nodes: 1000, Edges: 19817
Degree: min=25, max=53, mean=39.6, median=39.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	9.1288	57	273.864	168	357	1.00001
exanteMDAG	1000	31000	37.861	46	1173.69	720	1530	1.00002
expost	1000	30000	9.1287	57	273.861	168	357	1.00003
expostMDAG	1000	31000	37.861	46	1173.69	720	1530	1.00002



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	3.0	0.3	0.0	0.5	0.0	0.0	1.0
1.0	2.0	53.0	10.7	10.0	4.5	8.0	13.0	16.0
2.0	114.0	282.0	205.8	205.0	22.6	191.0	221.0	244.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	24.0	59.0	39.0	39.0	0.0	39.0	40.0	40.0
2.0	24.0	59.0	39.0	39.0	0.0	39.0	40.0	40.0
3.0	24.0	59.0	39.0	39.0	0.0	39.0	40.0	40.0
4.0	24.0	59.0	39.0	39.0	0.0	39.0	40.0	40.0
5.0	24.0	59.0	39.0	39.0	0.0	39.0	40.0	40.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	24.0	59.0	39.0	39.0	0.0	39.0	40.0	40.0
2.0	24.0	59.0	39.0	39.0	0.0	39.0	40.0	40.0
3.0	24.0	59.0	39.0	39.0	0.0	39.0	40.0	40.0
4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9593	0	1000	39.6	273.9
exanteMDAG	0.9593	0	1000	39.6	1173.7
expost	0.9593	0	1000	39.6	273.9
expostMDAG	0.9593	0	1000	39.6	1173.7

Committee Analysis:
Elections: 1000, Avg size: 7.0, Range: 7-7
Unique members: 7, Selections: 1000-1000

Similarity Analysis:
Nodes compared: 1000, Unique members: 7
Global intersection: 7, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:
Total alerts: 991
WARN: global: 991

Simulation Analysis: baseline-honest-no-drop-medium-degree-20251024-103431

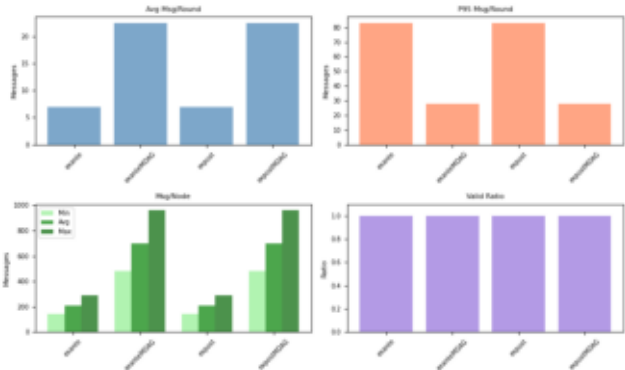
SIMULATION: baseline-honest-no-drop-medium-degree-20251024-103431
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Files: 1000, Records: 399257

Network Graph Analysis:
Nodes: 1000, Edges: 11931
Degree: min=16, max=42, mean=23.9, median=24.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	6.96953	83	209.086	144	288	1.00001
exanteMDAG	1000	31000	22.4817	28	696.933	480	960	1.00003
expost	1000	30000	6.9696	83	209.088	144	288	1
expostMDAG	1000	31000	22.4819	28	696.939	480	960	1.00002



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	2.0	0.2	0.0	0.5	0.0	0.0	1.0
1.0	0.0	35.0	4.9	4.0	3.1	3.0	6.0	9.0
2.0	43.0	141.0	84.2	84.0	14.7	73.0	94.0	111.0
3.0	72.0	214.0	119.3	106.0	17.8	107.0	132.0	150.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	16.0	32.0	23.0	23.0	0.0	20.0	26.0	28.0
2.0	16.0	32.0	23.0	23.0	0.0	20.0	26.0	28.0
3.0	16.0	32.0	23.0	23.0	0.0	20.0	26.0	28.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	16.0	32.0	23.2	23.0	0.0	20.0	26.0	28.0
2.0	16.0	32.0	23.0	23.0	0.0	20.0	26.0	28.0
3.0	16.0	32.0	23.0	23.0	0.0	20.0	26.0	28.0
4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9257	0	1000	23.9	209.1
exanteMDAG	0.9257	0	1000	23.9	696.9
expost	0.9257	0	1000	23.9	209.1
expostMDAG	0.9257	0	1000	23.9	696.9

Committee Analysis:
Elections: 1000, Avg size: 9.0, Range: 9-9
Unique members: 9, Selections: 1000-1000

Similarity Analysis:
Nodes compared: 1000, Unique members: 9
Global intersection: 9, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:
Total alerts: 986
WARN: global: 986

Simulation Analysis: baseline-honest-no-drop-small-degree-20251024-075646

SIMULATION: baseline-honest-no-drop-small-degree-20251024-075646

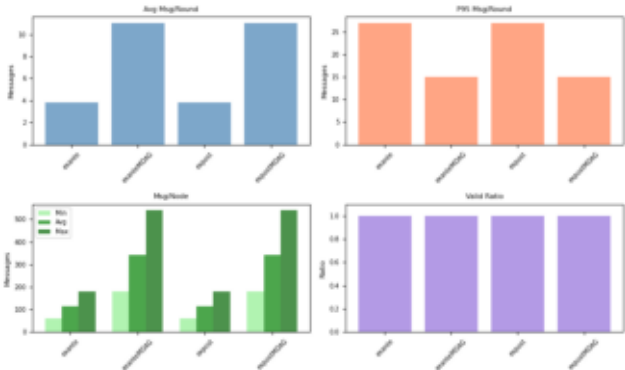
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Files: 1000, Records: 371602

Network Graph Analysis:
Nodes: 1000, Edges: 5984
Degree: min=6, max=23, mean=12.0, median=12.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	3.8	27	114	60	180	1
exanteMDAG	1000	31000	11.0318	15	341.987	180	540	1.00002
expost	1000	30000	3.7999	27	113.997	60	180	1.00003
expostMDAG	1000	31000	11.0317	15	341.982	180	540	1.00005



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	2.0	0.1	0.0	0.3	0.0	0.0	1.0
1.0	0.0	15.0	1.3	1.0	1.5	0.0	2.0	3.0
2.0	2.0	35.0	13.3	12.0	5.1	10.0	16.0	23.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4.0	8.0	62.0	26.8	26.0	7.4	21.0	32.0	40.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9023	0	1000	12	114
exanteMDAG	0.9024	0	1000	12	342
expost	0.9024	0	1000	12	114
expostMDAG	0.9023	0	1000	12	342

Committee Analysis:

Elections: 1000, Avg size: 10.0, Range: 10-10
Unique members: 10, Selections: 1000-1000

Similarity Analysis:

Nodes compared: 1000, Unique members: 10
Global intersection: 10, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 985
WARN: global: 985

Simulation Analysis: big-committee-with-drop-large-degree-20251024-151036

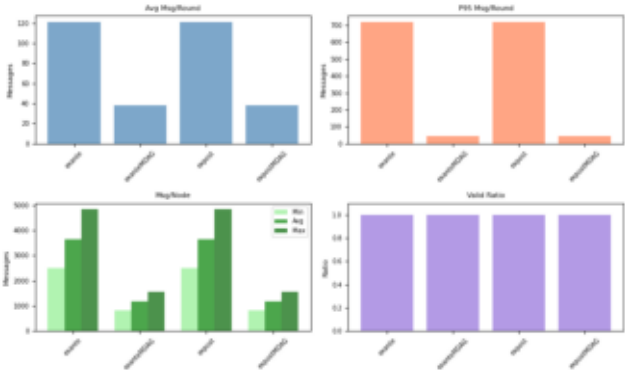
SIMULATION: big-committee-with-drop-large-degree-20251024-151036

Files: 1000, Records: 613543

Network Graph Analysis:
Nodes: 1000, Edges: 19803
Degree: min=27, max=56, mean=39.6, median=40.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	121.289	720.05	3638.67	2510	4836	1.00001
exanteMDAG	1000	31000	37.8625	46	1173.74	810	1560	1.00003
expost	1000	30000	121.291	720	3638.72	2509	4836	1.00001
expostMDAG	1000	31000	37.8633	46	1173.76	810	1560	1.00002



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	11.0	3.7	3.0	1.9	2.0	5.0	7.0
1.0	80.0	249.0	144.9	143.0	22.0	130.0	157.0	186.0
2.0	1885.0	3726.0	2771.1	2783.0	298.1	2561.8	2976.2	3241.1

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
2.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
3.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
4.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
2.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
3.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
4.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
5.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
6.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
7.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
8.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
9.0	2700.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9667	0	1000	39.6	3638.7
exanteMDAG	0.9666	0	1000	39.6	1173.7
expost	0.9666	0	1000	39.6	3638.7
expostMDAG	0.9666	0	1000	39.6	1173.8

Committee Analysis:

Elections: 1000, Avg size: 93.0, Range: 93-93
Unique members: 93, Selections: 1000-1000

Similarity Analysis:

Nodes compared: 1000, Unique members: 93
Global intersection: 93, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 3383
WARN: global: 990
WARN: 12D3KooWB4y8U5RfuTx2qwsayZEg49: 180
WARN: 12D3KooWSFjT9dDVAkvugTmihGz93h: 148

Simulation Analysis: big-committee-with-drop-medium-degree-20251024-123248

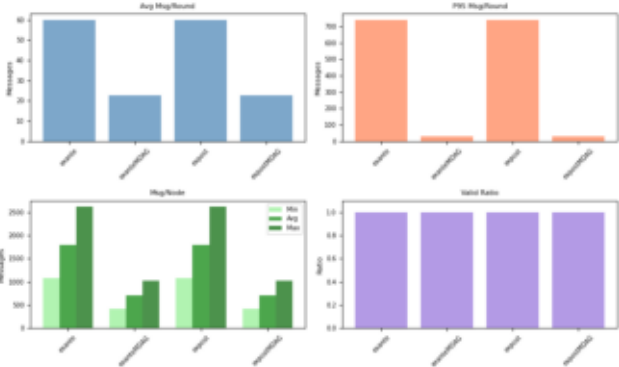
SIMULATION: big-committee-with-drop-medium-degree-20251024-123248
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Files: 1000, Records: 538393

Network Graph Analysis:
Nodes: 1000, Edges: 11932
Degree: min=15, max=36, mean=23.9, median=24.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	59.9087	739	1797.26	1077	2618	1.00001
exanteMDAG	1000	31000	22.5878	29	700.223	420	1020	1.00002
expost	1000	30000	59.9072	739.05	1797.22	1077	2618	1.00001
expostMDAG	1000	31000	22.5875	29	700.214	420	1020	1.00002



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	7.0	1.8	2.0	1.3	1.0	3.0	4.0
1.0	19.0	89.0	42.9	41.0	10.3	36.0	48.0	63.0
2.0	434.0	1117.0	742.6	742.0	107.6	667.0	813.2	927.0
3.0	611.0	1498.0	1010.0	1011.5	138.8	912.0	1095.8	1238.2

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	10.0	3400.0	2303.0	2300.0	302.0	2100.0	25.0.0	29.00.0
7.0	10.0	3400.0	2303.0	2300.0	302.0	2100.0	25.0.0	29.00.0
8.0	10.0	3400.0	2303.0	2300.0	302.0	2100.0	25.0.0	29.00.0
9.0	10.0	3400.0	2303.0	2300.0	302.0	2100.0	25.0.0	29.00.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
5.0	14.0	34.0	23.3	23.0	3.2	21.0	25.0	29.0
6.0	10.0	3470.0	2313.8	2320.0	312.3	2110.0	25.0.0	29.04.0
7.0	10.0	3490.0	2423.8	2410.0	302.3	2360.0	25.48.0	29.03.0
8.0	434.0	1345.0	742.35	742.0	107.25	666.0	819.0	29095.0
9.0	607.0	1499.0	1030.31	1031.0	138.28	912.0	1098.2	29241.0
4.0	0.0	2.0	0.0	0.0	0.3	0.0	0.0	0.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9483	0	1000	23.9	1797.3
exanteMDAG	0.9484	0	1000	23.9	700.2
expost	0.9483	0	1000	23.9	1797.2
expostMDAG	0.9483	0	1000	23.9	700.2

Committee Analysis:
Elections: 1000, Avg size: 77.0, Range: 77-77
Unique members: 77, Selections: 1000-1000

Similarity Analysis:
Nodes compared: 1000, Unique members: 77
Global intersection: 77, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:
Total alerts: 1959
WARN: global: 984
WARN: 12D3KooWDQ9BRGgvvYMYD3cnifXPxj: 164
WARN: 12D3KooWMR9CNShxAUBCZV98EsULue: 98

Simulation Analysis: big-committee-with-drop-small-degree-20251024-095503

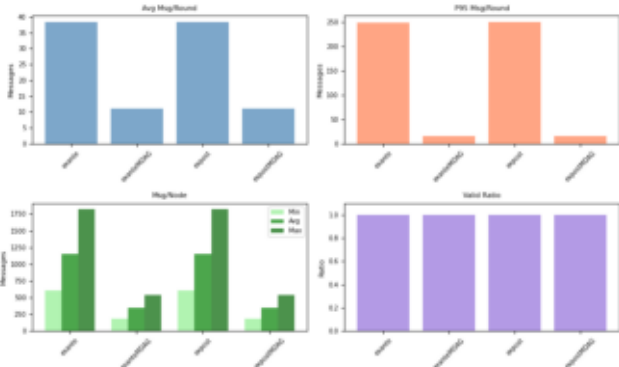
SIMULATION: big-committee-with-drop-small-degree-20251024-095503

Files: 1000, Records: 555785

Network Graph Analysis:
Nodes: 1000, Edges: 5985
Degree: min=6, max=24, mean=12.0, median=12.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	38.4221	249	1152.66	605	1818	1.00002
exanteMDAG	1000	31000	11.044	15	342.365	180	540	1.00003
expost	1000	30000	38.4216	250	1152.65	605	1817	1.00002
expostMDAG	1000	31000	11.0447	15	342.387	180	540	1.00003



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	6.0	1.2	1.0	1.0	0.0	2.0	3.0
1.0	2.0	42.0	14.0	13.0	5.5	10.8	16.0	25.0
2.0	73.0	258.0	142.1	141.0	32.7	120.0	163.0	198.0
3.0	373.0	746.0	746.0	746.0	145.3	649.0	834.2	993.1

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	1.0	0.0	0.0	0.1	0.0	0.0	0.0
1.0	0.0	18000	11040	11000	2020	10000	13000	15000
2.0	0.0	18000	11040	11000	2020	10000	13000	15000
3.0	0.0	18000	11040	11000	2020	10000	13000	15000

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	18.0	11.4	11.0	2.2	10.0	13.0	15.0
1.0	0.0	18000	11142	11100	2120	10000	13200	15300
2.0	0.0	18200	11440	11300	2525	10100	13600	15500
3.0	79.0	28800	14241	14100	3226	10000	10300	19800
4.0	379.0	128900	74548	74900	14524	60900	83602	99200
5.0	105.0	411.0	249.6	250.0	51.5	214.0	286.0	335.0
6.0	0.0	1.0	0.0	0.0	0.1	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9081	0	1000	12	1152.7
exanteMDAG	0.908	0	1000	12	342.4
expost	0.9081	0	1000	12	1152.6
expostMDAG	0.908	0	1000	12	342.4

Committee Analysis:

Elections: 1000, Avg size: 101.0, Range: 101-101
Unique members: 101, Selections: 1000-1000

Similarity Analysis:

Nodes compared: 1000, Unique members: 101
Global intersection: 101, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 1766
WARN: global: 989
WARN: 12D3KooWNgz79vTCNohB7JcjNERxDX: 148
WARN: 12D3KooWGrv8hnUAHztrsqidzffRCp: 130

Simulation Analysis: medium-committee-with-drop-large-degree-20251024-143108

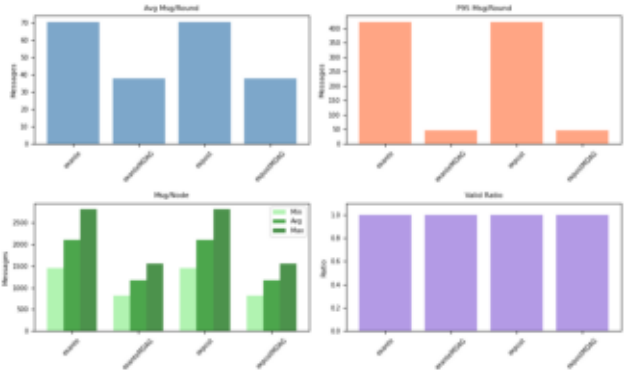
SIMULATION: medium-committee-with-drop-large-degree-20251024-143108
=====

Files: 1000, Records: 532576

Network Graph Analysis:
Nodes: 999, Edges: 19762
Degree: min=27, max=53, mean=39.6, median=39.0
Components: 1, Largest: 999

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	999	29970	70.263	421	2107.89	1458	2806	1.00001
exanteMDAG	999	30969	37.7748	46	1171.02	810	1560	1.00002
expost	999	29970	70.2627	421.55	2107.88	1458	2808	1.00001
expostMDAG	999	30969	37.7754	46	1171.04	810	1560	1.00003



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	7.0	2.1	2.0	1.4	1.0	3.0	5.0
1.0	51.0	152.0	83.7	82.0	14.9	74.0	91.0	110.1
2.0	1107.0	2147.0	1598.8	1599.0	172.2	1487.0	1720.0	1879.2
3.0	651.0	1578.0	463.3	461.0	50.7	388.5	457.0	510.1

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	270.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
2.0	270.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
3.0	270.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0
4.0	270.0	52.0	39.0	39.0	4.0	36.0	42.0	46.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	27.0	52.0	39.0	39.0	4.2	36.0	42.0	46.0
1.0	270.0	52.0	39.0	39.0	4.2	36.0	42.0	46.0
2.0	270.0	52.0	39.0	39.0	4.2	36.0	42.0	46.0
3.0	270.0	52.0	39.0	39.0	4.2	36.0	42.0	46.0
4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9656	0	999	39.6	2107.9
exanteMDAG	0.9656	0	999	39.6	1171
expost	0.9656	0	999	39.6	2107.9
expostMDAG	0.9656	0	999	39.6	1171

Committee Analysis:
Elections: 999, Avg size: 54.0, Range: 54-54
Unique members: 54, Selections: 999-999

Similarity Analysis:
Nodes compared: 999, Unique members: 54
Global intersection: 54, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:
Total alerts: 2628
WARN: global: 988
WARN: 12D3KooWLRkF2NDZ6Xg8bhFwqqiVLV: 200
WARN: 12D3KooWL7eT7rtVot8oQ3iL5sgnrg: 180

Simulation Analysis: medium-committee-with-drop-medium-degree-20251024-115320

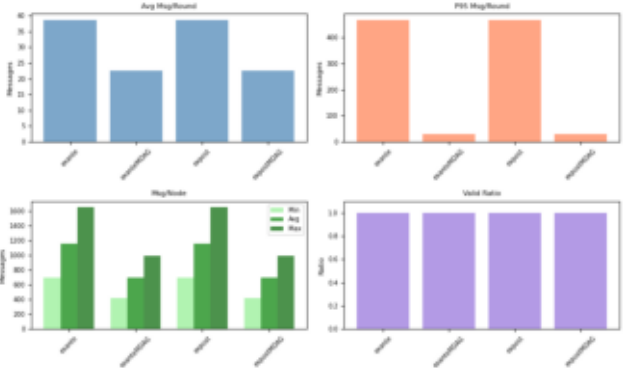
SIMULATION: medium-committee-with-drop-medium-degree-20251024-115320
=====

Files: 1000, Records: 484382

Network Graph Analysis:
Nodes: 1000, Edges: 11939
Degree: min=15, max=38, mean=23.9, median=24.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	38.687	468.05	1160.61	700	1650	1.00001
exanteMDAG	1000	31000	22.4639	28	696.38	420	990	1.00002
expost	1000	30000	38.6889	468	1160.67	698	1650	1.00001
expostMDAG	1000	31000	22.4634	28	696.366	419	990	1.00003



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	6.0	1.2	1.0	1.0	0.0	2.0	3.0
1.0	11.0	69.0	27.4	26.0	7.7	22.0	31.0	41.0
2.0	290.0	770.0	474.1	471.0	67.0	430.0	518.0	586.1
3.0	302.0	926.0	659.0	658.5	89.5	598.8	718.0	804.2
4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	10.0	30.0	20.0	20.0	0.0	20.0	20.0	20.0
7.0	10.0	30.0	20.0	20.0	0.0	20.0	20.0	20.0
8.0	10.0	30.0	20.0	20.0	0.0	20.0	20.0	20.0
9.0	10.0	30.0	20.0	20.0	0.0	20.0	20.0	20.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
2.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
3.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
4.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
5.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
6.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
7.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
8.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
9.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
2.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
3.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
4.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
5.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
6.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
7.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
8.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0
9.0	14.0	33.0	23.2	23.0	3.0	20.0	25.0	28.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9149	0	1000	23.9	1160.6
exanteMDAG	0.9151	0	1000	23.9	696.4
expost	0.9149	0	1000	23.9	1160.7
expostMDAG	0.9149	0	1000	23.9	696.4

Committee Analysis:
Elections: 1000, Avg size: 50.0, Range: 50-50
Unique members: 50, Selections: 1000-1000

Similarity Analysis:
Nodes compared: 1000, Unique members: 50
Global intersection: 50, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:
Total alerts: 2189
WARN: global: 985
WARN: 12D3KooWEzHeWhwjNDbRoBVg5htjtZ: 120
WARN: 12D3KooWBeQNU4wqdLioFe7NK6Ewbq: 118

Simulation Analysis: medium-committee-with-drop-small-degree-20251024-091537

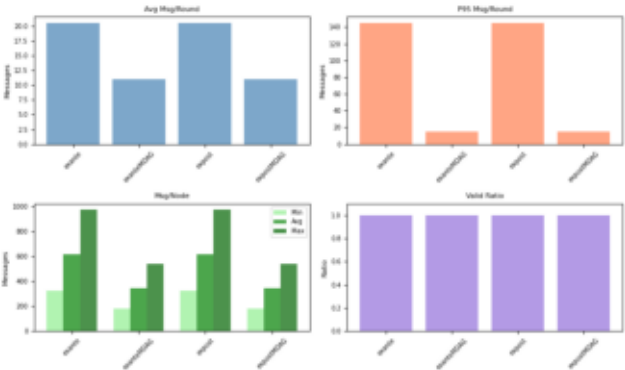
SIMULATION: medium-committee-with-drop-small-degree-20251024-091537

Files: 1000, Records: 461101

Network Graph Analysis:
Nodes: 1000, Edges: 5981
Degree: min=6, max=21, mean=12.0, median=12.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	20.5483	145	616.45	324	972	1.00002
exanteMDAG	1000	31000	11.0473	15	342.467	180	540	1.00004
expost	1000	30000	20.5493	145	616.478	323	972	1.00003
expostMDAG	1000	31000	11.0479	15	342.484	180	540	1.00004



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	4.0	0.6	0.0	0.7	0.0	1.0	2.0
1.0	0.0	25.0	7.1	6.0	3.7	5.0	9.0	14.0
2.0	30.0	133.0	72.4	71.0	17.8	59.8	84.0	103.0
3.0	60.0	300.0	100.0	100.0	78.2	339.0	442.0	520.0
4.0	66.0	235.0	145.7	145.0	29.4	126.0	166.0	194.0
5.0	0.0	1.0	0.0	0.0	0.1	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
1.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
2.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
3.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
4.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
5.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
6.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
7.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
8.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
9.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
1.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
2.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
3.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
4.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
5.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
6.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
7.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
8.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0
9.0	0.0	10.0	10.0	10.0	0.0	10.0	10.0	10.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9158	0	1000	12	616.5
exanteMDAG	0.9158	0	1000	12	342.5
expost	0.9159	0	1000	12	616.5
expostMDAG	0.9157	0	1000	12	342.5

Committee Analysis:

Elections: 1000, Avg size: 54.0, Range: 54-54
Unique members: 54, Selections: 1000-1000

Similarity Analysis:

Nodes compared: 1000, Unique members: 54
Global intersection: 54, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 1492
WARN: global: 987
WARN: 12D3KooWJT1nef8rsKi2dWkvbRzhAT: 66
WARN: 12D3KooWMrBWdareJW1bVaYVbYyYhr: 66

Simulation Analysis: simulation-20251018-181638-n1000-m12-d6-c20

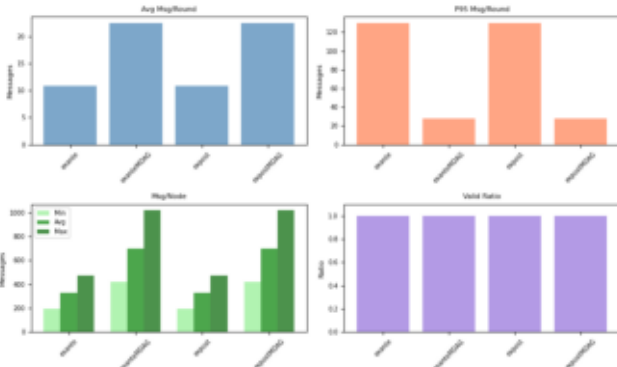
SIMULATION: simulation-20251018-181638-n1000-m12-d6-c20
=====

Files: 1001, Records: 408872

Network Graph Analysis:
Nodes: 999, Edges: 11916
Degree: min=15, max=37, mean=23.9, median=24.0
Components: 1, Largest: 999

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
-----	-----	-----	-----	-----	-----	-----	-----	-----
exante	999	29970	10.8626	130	325.878	196	476	1.00001
exanteMDAG	999	30969	22.5257	28	698.296	420	1020	1.00002
expost	999	29970	10.8627	130	325.88	196	476	1.00001
expostMDAG	999	30969	22.5259	28	698.302	420	1020	1.00002



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
-----	-----	-----	-----	-----	-----	-----	-----	-----
0.0	0.0	3.0	0.3	0.0	0.5	0.0	1.0	1.0
1.0	1.0	36.0	7.5	7.0	3.7	5.0	9.0	13.0
2.0	73.0	229.0	131.0	131.0	20.9	116.5	144.0	166.0
3.0	179.0	82.0	187.1	166.0	28.0	169.0	205.0	234.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
-----	-----	-----	-----	-----	-----	-----	-----	-----
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	10.0	30.0	20.0	20.0	0.0	20.0	20.0	20.0
7.0	10.0	30.0	20.0	20.0	0.0	20.0	20.0	20.0
8.0	10.0	30.0	20.0	20.0	0.0	20.0	20.0	20.0
9.0	10.0	30.0	20.0	20.0	0.0	20.0	20.0	20.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
-----	-----	-----	-----	-----	-----	-----	-----	-----
5.0	14.0	34.0	23.3	23.0	3.2	21.0	25.0	28.1
6.0	14.0	34.0	20.3	20.0	0.3	20.0	25.0	28.0
7.0	14.0	30.0	27.0	27.0	3.7	25.0	29.0	29.0
8.0	74.0	220.0	133.0	133.0	20.0	120.0	144.0	160.0
9.0	104.0	282.0	187.1	180.0	28.0	160.0	205.0	234.0
4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
-----	-----	-----	-----	-----	-----
exante	0.9331	0	999	23.9	325.9
exanteMDAG	0.9331	0	999	23.9	698.3
expost	0.9331	0	999	23.9	325.9
expostMDAG	0.9331	0	999	23.9	698.3

Committee Analysis:

Elections: 999, Avg size: 14.0, Range: 14-14
Unique members: 14, Selections: 999-999

Similarity Analysis:

Nodes compared: 999, Unique members: 14
Global intersection: 14, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 993
WARN: global: 993

Simulation Analysis: simulation-20251018-185604-n1000-m12-d6-c20

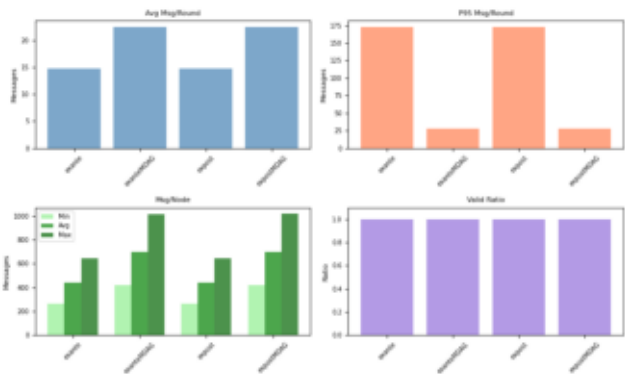
SIMULATION: simulation-20251018-185604-n1000-m12-d6-c20
=====

Files: 1001, Records: 420564

Network Graph Analysis:
Nodes: 1000, Edges: 11936
Degree: min=15, max=38, mean=23.9, median=24.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	14.7264	173	441.792	266	646	1.00001
exanteMDAG	1000	31000	22.5009	28	697.528	420	1019	1.00002
expost	1000	30000	14.725	173	441.751	266	646	1.00001
expostMDAG	1000	31000	22.5006	28	697.519	420	1020	1.00003



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	4.0	0.4	0.0	0.6	0.0	1.0	2.0
1.0	1.0	37.0	9.9	9.0	4.2	7.0	12.0	16.0
2.0	101.0	271.0	173.8	173.0	26.9	154.0	193.0	219.0
3.0	169.0	289.0	257.6	257.0	33.7	233.0	283.0	319.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	10.0	30.0	29.0	29.0	0.0	20.0	20.0	20.0
2.0	10.0	30.0	29.0	29.0	0.0	20.0	20.0	20.0
3.0	10.0	30.0	29.0	29.0	0.0	20.0	20.0	20.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	14.0	34.0	23.3	23.0	3.2	21.0	25.0	28.0
2.0	14.0	34.0	23.3	23.0	0.0	20.0	25.0	28.0
3.0	14.0	34.0	23.3	23.0	0.0	20.0	25.0	28.0
4.0	14.0	34.0	23.3	23.0	0.0	20.0	25.0	28.0
5.0	14.0	34.0	23.3	23.0	0.0	20.0	25.0	28.0
6.0	14.0	34.0	23.3	23.0	0.0	20.0	25.0	28.0
7.0	14.0	34.0	23.3	23.0	0.0	20.0	25.0	28.0
8.0	14.0	34.0	23.3	23.0	0.0	20.0	25.0	28.0
9.0	14.0	34.0	23.3	23.0	0.0	20.0	25.0	28.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9328	0	1000	23.9	441.8
exanteMDAG	0.9327	0	1000	23.9	697.5
expost	0.9327	0	1000	23.9	441.8
expostMDAG	0.9328	0	1000	23.9	697.5

Committee Analysis:
Elections: 1000, Avg size: 19.0, Range: 19-19
Unique members: 19, Selections: 1000-1000

Similarity Analysis:
Nodes compared: 1000, Unique members: 19
Global intersection: 19, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:
Total alerts: 1344
WARN: global: 991
WARN: 12D3KooWD4mm9e2rUwrdbBrGRRrA9p: 44
WARN: 12D3KooWDgLy2MAUxaEjGvGcTXUixQ: 30

Simulation Analysis: simulation-20251018-201456-n1000-m12-d6-c50

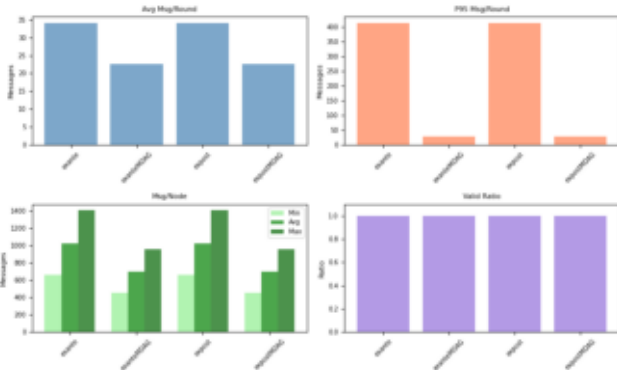
SIMULATION: simulation-20251018-201456-n1000-m12-d6-c50
=====

Files: 1001, Records: 468821

Network Graph Analysis:
Nodes: 999, Edges: 11918
Degree: min=15, max=34, mean=23.9, median=24.0
Components: 1, Largest: 999

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	999	29970	34.1251	414	1023.75	660	1408	1.00001
exanteMDAG	999	30969	22.5163	28	698.004	450	960	1.00001
expost	999	29970	34.1251	414	1023.75	660	1408	1.00001
expostMDAG	999	30969	22.5161	28	697.998	450	960	1.00002



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	5.0	1.0	1.0	1.0	0.0	2.0	3.0
1.0	8.0	66.0	24.1	23.0	7.3	19.0	27.0	36.0
2.0	259.0	610.0	418.0	419.0	60.4	375.0	460.0	520.0
3.0	379.0	72.0	580.6	580.0	79.9	525.0	635.0	718.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	19.0	39.0	29.0	29.0	0.0	29.0	29.0	29.0
2.0	19.0	39.0	29.0	29.0	0.0	29.0	29.0	29.0
3.0	19.0	39.0	29.0	29.0	0.0	29.0	29.0	29.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	3.0	1.0	1.0	1.0	0.0	2.0	3.0
1.0	8.0	66.0	24.1	23.0	7.3	19.0	27.0	36.0
2.0	259.0	610.0	418.0	419.0	60.4	375.0	460.0	520.0
3.0	379.0	72.0	580.6	580.0	79.9	525.0	635.0	718.0

4.0	0.0	2.0	0.1	0.0	0.2	0.0	0.0	1.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9413	0	999	23.9	1023.8
exanteMDAG	0.9413	0	999	23.9	698
expost	0.9413	0	999	23.9	1023.8
expostMDAG	0.9413	0	999	23.9	698

Committee Analysis:

Elections: 999, Avg size: 44.0, Range: 44-44
Unique members: 44, Selections: 999-999

Similarity Analysis:

Nodes compared: 999, Unique members: 44
Global intersection: 44, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 993
WARN: global: 993

Simulation Analysis: simulation-20251018-205424-n1000-m12-d6-c50

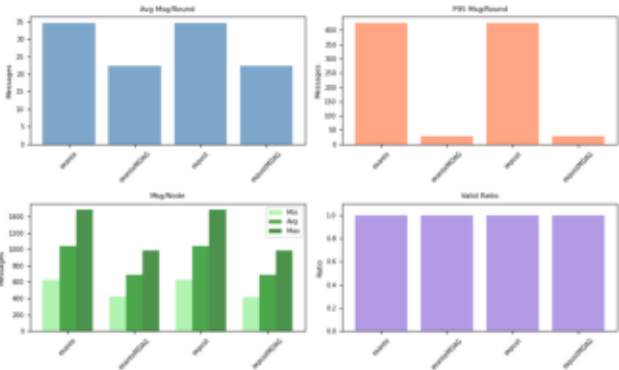
SIMULATION: simulation-20251018-205424-n1000-m12-d6-c50
=====

Files: 1001, Records: 473790

Network Graph Analysis:
Nodes: 999, Edges: 11912
Degree: min=14, max=36, mean=23.8, median=24.0
Components: 1, Largest: 999

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	999	29970	34.6533	426	1039.6	630	1485	1.00001
exanteMDAG	999	30969	22.3569	28	693.063	420	990	1.00001
expost	999	29970	34.6519	426	1039.56	630	1485	1.00001
expostMDAG	999	30969	22.3553	28	693.013	419	990	1.00003



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	5.0	1.1	1.0	1.0	0.0	2.0	3.0
1.0	5.0	64.0	24.7	24.0	7.3	20.0	28.0	37.0
2.0	247.0	621.0	428.0	429.0	61.0	386.0	468.5	534.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	0.0	39.0	29.0	29.0	0.0	29.0	29.0	29.0
2.0	0.0	39.0	29.0	29.0	0.0	29.0	29.0	29.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	3.0	2.0	2.0	0.0	2.0	2.0	2.0
1.0	14.0	64.0	24.7	24.0	7.3	20.0	28.0	37.0
2.0	247.0	624.0	428.0	429.0	61.0	387.0	469.0	534.0
3.0	352.0	849.0	589.8	588.0	88.3	533.0	639.0	717.0
4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9168	0	999	23.8	1039.6
exanteMDAG	0.9168	0	999	23.8	693.1
expost	0.9168	0	999	23.8	1039.6
expostMDAG	0.9167	0	999	23.8	693

Committee Analysis:

Elections: 999, Avg size: 45.0, Range: 45-45
Unique members: 45, Selections: 999-999

Similarity Analysis:

Nodes compared: 999, Unique members: 45
Global intersection: 45, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 2224
WARN: global: 990
WARN: 12D3KooWEHqa1WXHA3QCTP7of5D2j5: 128
WARN: 12D3KooWBGbDGP7ALPSuC8Ea7EdjpF: 116

Simulation Analysis: simulation-20251018-221318-n1000-m12-d6-c100

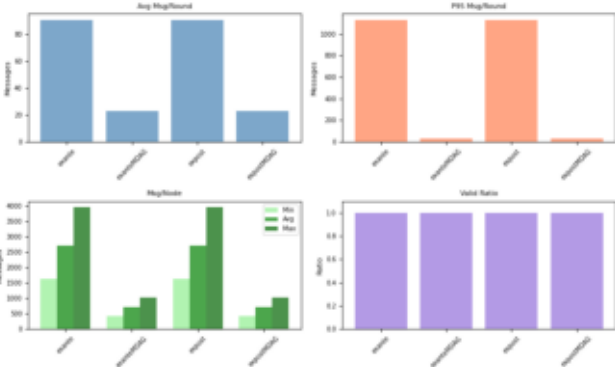
SIMULATION: simulation-20251018-221318-n1000-m12-d6-c100
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Files: 1001, Records: 612802

Network Graph Analysis:
Nodes: 999, Edges: 11933
Degree: min=14, max=37, mean=23.9, median=24.0
Components: 1, Largest: 999

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	999	29970	90.3918	1131	2711.75	1624	3944	1.00001
exanteMDAG	999	30969	22.6225	29	701.298	420	1020	1.00003
expost	999	29970	90.3919	1131	2711.76	1624	3944	1.00001
expostMDAG	999	30969	22.6228	29	701.308	420	1020	1.00002



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	9.0	2.8	3.0	1.6	2.0	4.0	6.0
1.0	31.0	126.0	66.2	65.0	13.9	57.0	74.0	92.0
2.0	689.0	1743.0	1142.3	1141.0	159.0	1039.5	1241.5	1412.2
3.0	899.0	2143.0	1499.0	1499.0	201.2	1368.0	1635.0	1827.2

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	10.0	3490.0	2304.0	2300.0	301.0	21.0	25.0	0.0
7.0	10.0	3490.0	2304.0	2300.0	301.0	21.0	25.0	0.0
8.0	10.0	3490.0	2304.0	2300.0	301.0	21.0	25.0	0.0
9.0	10.0	3490.0	2304.0	2300.0	301.0	21.0	25.0	0.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
5.0	14.0	34.0	23.4	23.0	3.1	21.0	25.0	29.0
6.0	10.0	3490.0	2324.8	2330.0	311.6	21.0	25.0	0.0
7.0	34.0	3490.0	2064.2	2050.0	331.9	21.0	25.0	0.0
8.0	689.0	1743.0	1142.3	1142.0	159.0	1039.5	1241.5	1412.2
9.0	899.0	2143.0	1499.0	1499.0	201.2	1368.0	1635.0	1827.2
4.0	0.0	5.0	0.6	0.0	0.8	0.0	1.0	2.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9539	0	999	23.9	2711.8
exanteMDAG	0.9539	0	999	23.9	701.3
expost	0.9539	0	999	23.9	2711.8
expostMDAG	0.954	0	999	23.9	701.3

Committee Analysis:

Elections: 999, Avg size: 116.0, Range: 116-116
Unique members: 116, Selections: 999-999

Similarity Analysis:

Nodes compared: 999, Unique members: 116
Global intersection: 116, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 992
WARN: global: 992

Simulation Analysis: simulation-20251018-225249-n1000-m12-d6-c100

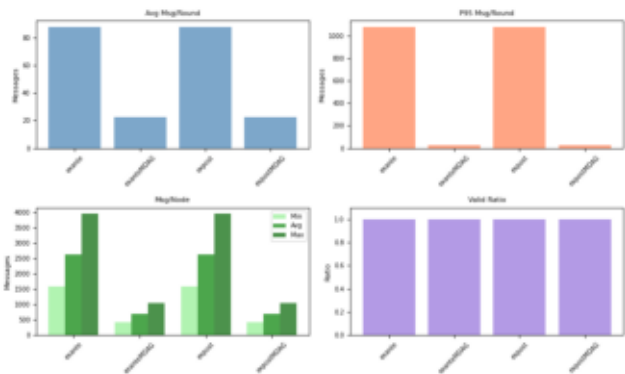
SIMULATION: simulation-20251018-225249-n1000-m12-d6-c100
=====

Files: 1001, Records: 613100

Network Graph Analysis:
Nodes: 1000, Edges: 11929
Degree: min=14, max=37, mean=23.9, median=24.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	87.8486	1079	2635.46	1582	3954	1.00001
exanteMDAG	1000	31000	22.57	28	699.669	420	1050	1.00002
expost	1000	30000	87.8494	1079	2635.48	1582	3953	1.00001
expostMDAG	1000	31000	22.5703	28	699.68	420	1050	1.00003



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	9.0	2.6	3.0	1.6	1.0	4.0	5.0
1.0	28.0	118.0	63.0	62.0	13.4	54.0	70.0	87.0
2.0	647.0	1674.0	1086.5	1090.0	155.5	983.0	1185.2	1341.1
3.0	865.0	2109.0	1483.7	1485.0	201.6	1343.5	1618.5	1810.0
4.0	865.0	2109.0	1483.7	1485.0	201.6	1343.5	1618.5	1810.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	10.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
7.0	10.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
8.0	10.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
9.0	10.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	10.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
2.0	10.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
3.0	10.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
4.0	10.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
5.0	10.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	14.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
2.0	14.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
3.0	14.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
4.0	14.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
5.0	14.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
6.0	14.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
7.0	14.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
8.0	14.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0
9.0	14.0	3500.0	2303.0	2300.0	302.0	21.0	25.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9487	0	1000	23.9	2635.5
exanteMDAG	0.9487	0	1000	23.9	699.7
expost	0.9487	0	1000	23.9	2635.5
expostMDAG	0.9487	0	1000	23.9	699.7

Committee Analysis:

Elections: 1000, Avg size: 113.0, Range: 113-113
Unique members: 113, Selections: 1000-1000

Similarity Analysis:

Nodes compared: 1000, Unique members: 113
Global intersection: 113, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 3201
WARN: global: 995
WARN: 12D3KooWPq42pEyXJzAuqJ8a7sDAP4: 146
WARN: 12D3KooWL9YUqdhUnE77MGP2JckVbV: 132

Simulation Analysis: small-committee-with-drop-large-degree-20251024-135142

SIMULATION: small-committee-with-drop-large-degree-20251024-135142

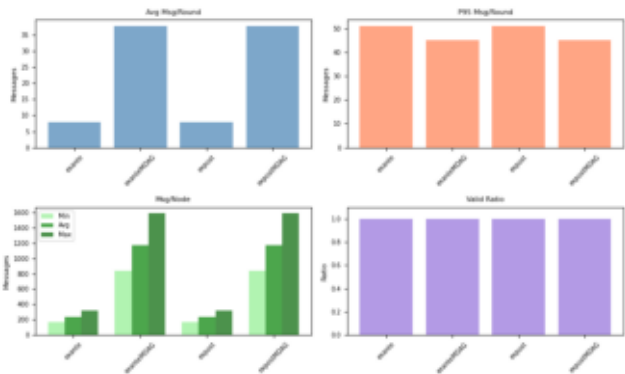
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Files: 1000, Records: 433693

Network Graph Analysis:
Nodes: 1000, Edges: 19784
Degree: min=28, max=53, mean=39.6, median=39.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	7.80833	51	234.25	168	318	1.00003
exanteMDAG	1000	31000	37.7822	45	1171.25	837	1589	1.00002
expost	1000	30000	7.80863	51	234.259	168	318	1
expostMDAG	1000	31000	37.7819	45.05	1171.24	839	1589	1.00002



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	2.0	0.2	0.0	0.5	0.0	0.0	1.0
1.0	1.0	49.0	8.9	8.0	3.8	7.0	10.0	14.0
2.0	116.0	235.0	174.1	175.0	18.6	161.0	186.0	205.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
2.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
3.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
4.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
5.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
2.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
3.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
4.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
5.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
6.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
7.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
8.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0
9.0	28.0	59.0	39.0	39.0	9.0	38.0	49.0	49.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9594	0	1000	39.6	234.2
exanteMDAG	0.9592	0	1000	39.6	1171.2
expost	0.9591	0	1000	39.6	234.3
expostMDAG	0.9592	0	1000	39.6	1171.2

Committee Analysis:

Elections: 1000, Avg size: 6.0, Range: 6-6
Unique members: 6, Selections: 1000-1000

Similarity Analysis:

Nodes compared: 1000, Unique members: 6
Global intersection: 6, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 1224
WARN: global: 978
WARN: 12D3KooWKmKGGZmFJ4RocngH8TehkG: 22
WARN: 12D3KooWBwCSCqBe9DHBVokpnxraex: 18

Simulation Analysis: small-committee-with-drop-medium-degree-20251024-111356

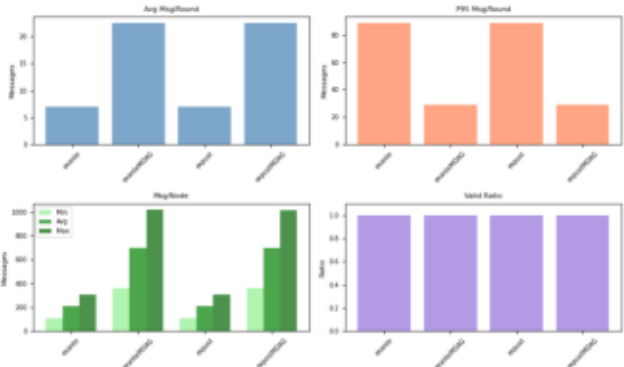
SIMULATION: small-committee-with-drop-medium-degree-20251024-111356
=====

Files: 1000, Records: 400049

Network Graph Analysis:
Nodes: 1000, Edges: 11939
Degree: min=13, max=37, mean=23.9, median=24.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	6.99523	89	209.857	108	306	1.00002
exanteMDAG	1000	31000	22.5663	29	699.556	360	1020	1.00003
expost	1000	30000	6.99543	89	209.863	108	305	1
expostMDAG	1000	31000	22.5663	29	699.554	360	1019	1.00002



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	2.0	0.2	0.0	0.5	0.0	0.0	1.0
1.0	0.0	36.0	5.5	5.0	3.3	4.0	7.0	10.0
2.0	43.0	151.0	93.2	92.0	15.8	82.0	104.0	119.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	0.0	34.0	29.0	29.0	0.0	29.0	29.0	29.0
2.0	0.0	34.0	29.0	29.0	0.0	29.0	29.0	29.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.0	0.0	34.0	29.0	29.0	0.0	29.0	29.0	29.0
2.0	0.0	34.0	29.0	29.0	0.0	29.0	29.0	29.0
3.0	0.0	34.0	29.0	29.0	0.0	29.0	29.0	29.0
4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.9468	0	1000	23.9	209.9
exanteMDAG	0.9469	0	1000	23.9	699.6
expost	0.9469	0	1000	23.9	209.9
expostMDAG	0.9469	0	1000	23.9	699.6

Committee Analysis:

Elections: 1000, Avg size: 9.0, Range: 9-9
Unique members: 9, Selections: 1000-1000

Similarity Analysis:

Nodes compared: 1000, Unique members: 9
Global intersection: 9, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 1064
WARN: global: 984
WARN: 12D3KooWDsNeYtjtkkRemNXFysFz7i: 14
WARN: 12D3KooWELBzhATxhhLUmgCbohPo9j: 14

Simulation Analysis: small-committee-with-drop-small-degree-20251024-083612

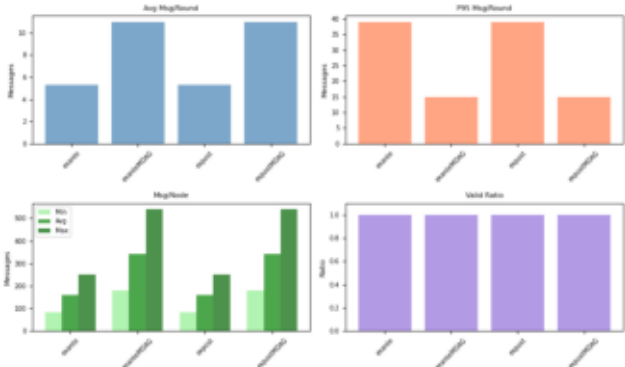
SIMULATION: small-committee-with-drop-small-degree-20251024-083612

Files: 1000, Records: 380087

Network Graph Analysis:
Nodes: 1000, Edges: 5983
Degree: min=7, max=21, mean=12.0, median=12.0
Components: 1, Largest: 1000

Protocol Message Statistics:

Protocol	Nodes	Total Rounds	Avg Msg/Round	P95 Msg/Round	Avg Msg/Node	Min Msg/Node	Max Msg/Node	Valid Ratio
exante	1000	30000	5.29873	39	158.962	84	252	1.00002
exanteMDAG	1000	31000	10.9878	15	340.621	180	540	1.00004
expost	1000	30000	5.2988	39	158.964	84	252	1.00001
expostMDAG	1000	31000	10.9877	15	340.62	179	540	1.00006



exante - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	2.0	0.2	0.0	0.4	0.0	0.0	1.0
1.0	0.0	18.0	1.8	1.0	1.8	1.0	2.0	4.0
2.0	4.0	52.0	18.2	18.0	6.4	14.0	22.0	30.0
3.0	4.0	65.0	19.3	19.0	6.5	85.0	112.0	132.0

exanteMDAG - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	18.0	19.0	19.0	0.0	10.0	19.0	19.0
1.0	0.0	18.0	19.4	19.0	0.0	10.0	19.0	19.0
2.0	0.0	18.0	19.4	19.0	0.0	10.0	19.0	19.0
3.0	0.0	18.0	19.4	19.0	0.0	10.0	19.0	19.0

expost - Per-Round Stats:

Round	Min	Max	Mean	Median	Std	P25	P75	P95
0.0	0.0	18.0	11.4	11.0	2.1	10.0	13.0	15.0
1.0	0.0	18.0	11.8	11.0	2.0	10.0	12.0	14.0
2.0	0.0	52.0	18.2	17.0	6.4	10.0	22.0	30.0
3.0	49.0	169.0	19.3	19.0	19.3	89.0	112.0	132.0
4.0	14.0	74.0	39.5	39.0	10.0	32.0	46.0	56.0
5.0	0.0	1.0	0.0	0.0	0.1	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Degree-Message Correlation:

(Pearson correlation between node degree and total messages)

Protocol	Correlation	P-value	Sample	Avg Degree	Avg Msgs
exante	0.8974	0	1000	12	159
exanteMDAG	0.8975	0	1000	12	340.6
expost	0.8977	0	1000	12	159
expostMDAG	0.8975	0	1000	12	340.6

Committee Analysis:

Elections: 1000, Avg size: 14.0, Range: 14-14
Unique members: 14, Selections: 1000-1000

Similarity Analysis:

Nodes compared: 1000, Unique members: 14
Global intersection: 14, Full consensus: True
Avg Jaccard: 1.000, Dice: 1.000, Overlap: 1.000

Error & Warning Audit:

Total alerts: 1089
WARN: global: 968
WARN: 12D3KooWSYgo3hTdjtHPsJrPfuaR94: 26
WARN: 12D3KooWCiBy5CJnfQLfyqWgzUHMh: 22